Introduction

Trajectory inference orders single cells along a continuous developmental or activation path, producing a pseudotime value per cell that represents its progress along the trajectory. Biological processes such as cell differentiation, cell-cycle progression, T-cell activation, and disease progression are continuous in nature; clustering imposes discrete categories that conceal the underlying continuous structure. Trajectory methods recover that continuity by fitting smooth curves or graphs through the data in dimensional-reduced space.

Prerequisites

A working understanding of scRNA-seq normalisation, dimensional reduction (PCA, UMAP), clustering, and the conceptual difference between discrete cell types and continuous developmental trajectories.

Theory

The two dominant trajectory frameworks:

Slingshot constructs a minimum spanning tree on cluster centroids in a reduced-dimensional space (typically PCA), specifies or infers a starting cluster, and fits smooth principal curves through each lineage from the start. Pseudotime is the arc length along each curve, computed per cell.

Monocle3 uses a reverse-graph-embedding approach on UMAP coordinates, fitting a principal graph and ordering cells along it. Monocle3 handles complex branching topologies more naturally than Slingshot.

Both methods require either a specified root cluster or an inferred one (with biological-marker-based validation strongly recommended). Pseudotime values are then used as a continuous covariate in downstream gene-by-pseudotime models (`tradeSeq`, generalised additive models) to identify genes whose expression varies along the trajectory.

Assumptions

Cells lie on a low-dimensional biological manifold; the cluster topology approximates the trajectory topology; the chosen root cluster represents the developmental starting state.

R Implementation

```
library(slingshot); library(SingleCellExperiment)

set.seed(2026)
n_cells <- 200
x <- seq(0, 1, length.out = n_cells)
pc1 <- x + rnorm(n_cells, 0, 0.05)
pc2 <- x^2 + rnorm(n_cells, 0, 0.05)

sce <- SingleCellExperiment(
  assays = list(counts = matrix(rpois(n_cells * 50, 3), 50, n_cells))
)
reducedDim(sce, "PCA") <- cbind(pc1, pc2)
sce$cluster <- factor(cut(x, breaks = 4, labels = 1:4))

sce <- slingshot(sce, clusterLabels = sce$cluster,
                 reducedDim = "PCA", start.clus = "1")
head(slingPseudotime(sce))

plot(reducedDim(sce, "PCA"), col = sce$cluster, pch = 16, asp = 1,
```

```
    main = "Slingshot lineage")
lines(SlingshotDataSet(sce), lwd = 2, col = "black")
```

Output & Results

`slingshot()` returns the `SingleCellExperiment` object augmented with pseudotime values (one column per lineage). The plot overlays smooth principal curves on the PCA scatter, with cells coloured by cluster, making the inferred lineage structure visible.

Interpretation

A reporting sentence: "Slingshot trajectory inference on the cluster-resolved PCA embedding recovered a single lineage from cluster 1 (root, validated as quiescent stem cells by SOX2 expression) through clusters 2 and 3 to cluster 4 (terminally activated cells); pseudotime correlated strongly with the known maturation marker BMI1 ($\rho = 0.84$, $p < 0.001$)." Always validate trajectory direction with known biology.

Practical Tips

- Specify `start.clus` explicitly when biology supports a clear initial state; automatic root inference is often unreliable for complex datasets.
- Compute trajectories on PCA space, not UMAP; UMAP distorts distances and produces unstable trajectory inference.
- Test genes for differential expression along pseudotime with `tradeSeq` (generalised additive models on count data) for principled identification of trajectory-driven genes.
- Monocle3 is preferred when the trajectory has complex branching structure (multiple lineages diverging); Slingshot is simpler and works well for linear or weakly branching trajectories.
- Always validate the inferred direction against known biological markers (start-state vs end-state); pseudotime is direction-less without a defined root.
- Pair trajectory analysis with cluster annotation; the lineage structure should connect biologically meaningful states, not arbitrary clusters.

R Packages Used

`slingshot` for principal-curve-based trajectory inference; `monocle3` for graph-based trajectory inference with branching support; `tradeSeq` for differential gene expression along pseudotime; `condiments` for trajectory testing under multiple conditions; `dyno` for unified access to dozens of trajectory inference methods.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- RNA-seq with DESeq2 / edgeR
- Enrichment analysis (GSEA / ORA)
- scRNA-seq with Seurat